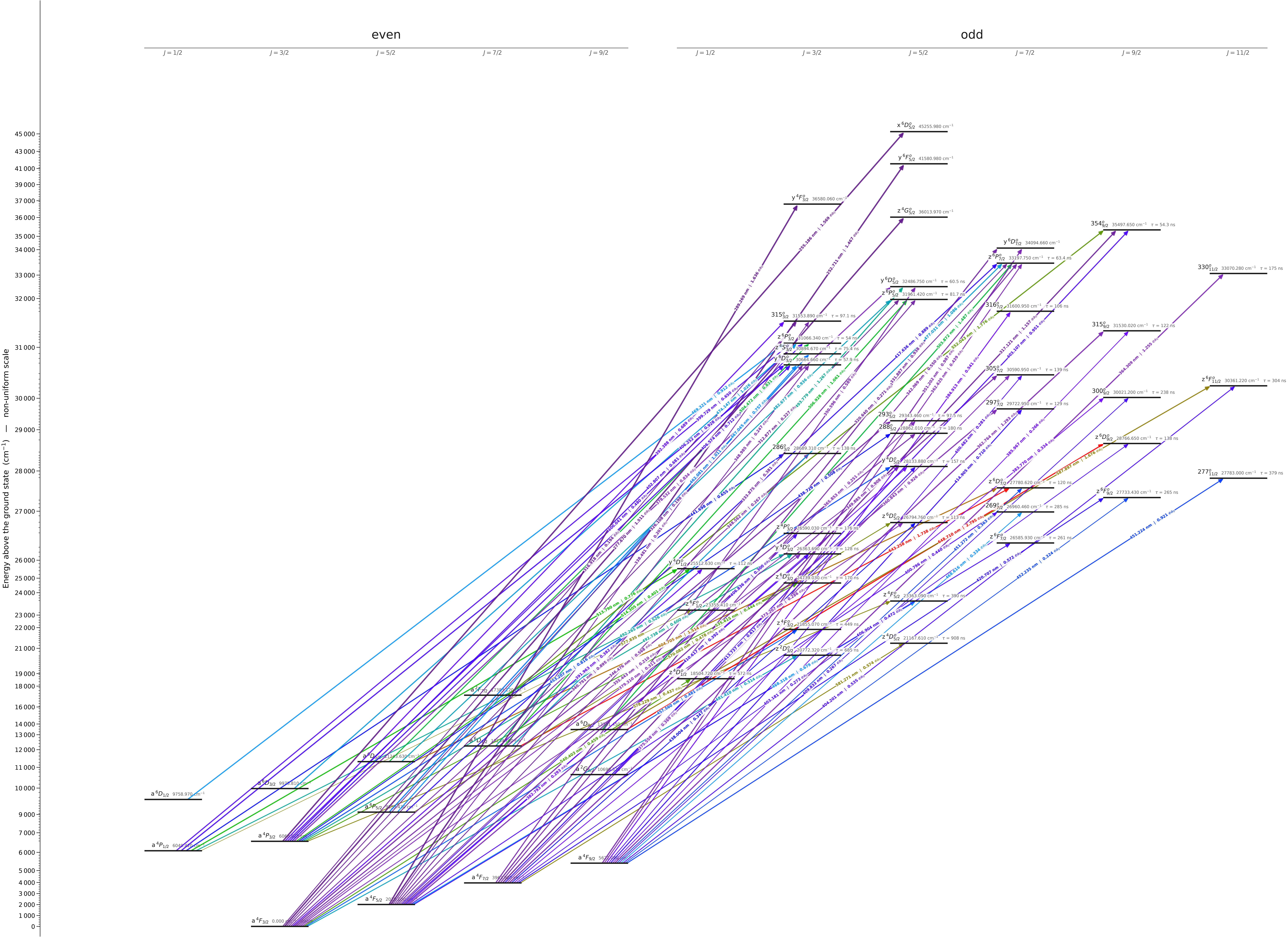
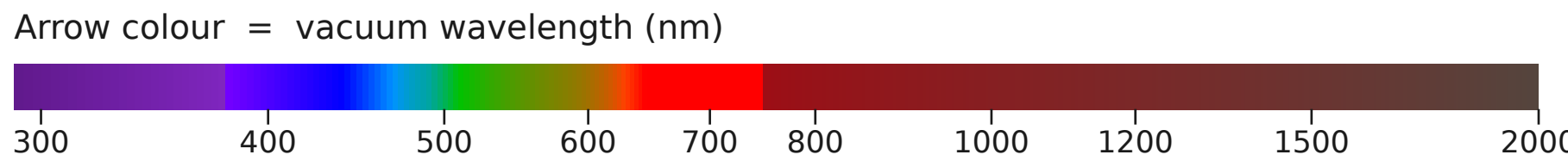


# Tantalum

Ta (Ta I). Strongest classified transitions below 2 μm (NIST): 55 levels, 90 lines.

Ta+ ionisation limit 60891.40 cm<sup>-1</sup> = 7.54957 eV (164.227 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle J||e r||J'\rangle$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model or semi-empirical calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                   | λ vac (nm) | λ air (nm) | ν (THz)  | $\langle J  e r  J'\rangle$ ( $e a_0$ ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                   |
|------------------------------|------------|------------|----------|-----------------------------------------|----------------------|----------------|------------|---------------------------------------|
| $a^4P_{5/2} - z^4D_{5/2}^o$  | 570.0822   | 569.9241   | 525.8758 | 0.478                                   | 4.167e+05            | 1.408          | 4.6        | hyperfine-structure spectroscopy line |
| $a^4P_{1/2} - z^4F_{1/2}^o$  | 577.8346   | 577.6744   | 518.8205 | -                                       | -                    | -              | -          | hyperfine-structure spectroscopy line |
| $a^4P_{3/2} - z^4F_{3/2}^o$  | 578.2292   | 578.0689   | 518.4665 | 0.437                                   | 3.333e+05            | 0.4081         | 13         | hyperfine-structure spectroscopy line |
| $a^6D_{9/2} - z^6F_{11/2}^o$ | 587.8974   | 587.7345   | 509.9401 | 1.676                                   | 2.333e+06            | 0.5235         | 72         | hyperfine-structure spectroscopy line |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: Measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>181</sup>Ta hyperfine structure (I = 7/2)

| Level          | A (MHz)   | B (MHz)   | Shift of each F level from the centre of gravity (MHz)                                                              |
|----------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------|
| $a^4P_{1/2}$   | 879(10)   | -         | F=3: -1977.75 F=4: +1538.25                                                                                         |
| $a^4P_{3/2}$   | 378(10)   | -1349(20) | F=2: -3274.18 F=3: -1176.61 F=4: +720.82 F=5: +1647.25                                                              |
| $a^4P_{5/2}$   | 251(10)   | 1670(20)  | F=1: -1929.11 F=2: -1904.25 F=3: -1688.04 F=4: -1065.75 F=5: +248.89 F=6: +2613.75                                  |
| $a^6D_{9/2}$   | 1089(10)  | -         | F=1: -20963.25 F=2: -18785.25 F=3: -15518.25 F=4: -11162.25 F=5: -5717.25 F=6: +816.75 F=7: +8439.75 F=8: +17151.75 |
| $z^6F_{1/2}^o$ | -1771(10) | -         | F=3: +3984.75 F=4: -3099.25                                                                                         |
| $z^4F_{5/2}^o$ | 480(10)   | -19(20)   | F=1: -5410.18 F=2: -4444.75 F=3: -2998.64 F=4: -1074.30 F=5: +1325.02 F=6: +4195.25                                 |
| $z^6D_{5/2}^o$ | 1033(10)  | 688(20)   | F=1: -11252.68 F=2: -9383.25 F=3: -6505.39 F=4: -2530.65 F=5: +2658.92 F=6: +9210.75                                |
| $z^6F_{3/2}^o$ | 669(10)   | 60(20)    | F=2: -15194.93 F=3: -13197.75 F=4: -10532.66 F=5: -7197.79 F=6: -3190.80 F=7: +1491.11 F=8: +6851.21 F=9: +12893.25 |

Measured A, B constants; sources in the data file.